Mat2002 Tutorial 10

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Foundamental matrix

Definition

Suppose $\vec{x}^{(1)}(t), \cdots, \vec{x}^{(n)}(t)$ form a fundamental set of solutions of

$$\vec{x}' = P(t)\vec{x}, \quad t \in I.$$

then the matrix

$$\Psi(t) = \begin{pmatrix} x_1^{(1)} & \cdots & x_1^{(n)} \\ \vdots & \cdots & \vdots \\ x_n^{(1)} & \cdots & x_n^{(n)} \end{pmatrix} = \left(\vec{x}^{(1)}, \cdots, \vec{x}^{(n)}\right)$$

is called the fundamental matrix for the linear system.

Remark

1). The general solution is

$$\vec{x} = c_1\vec{x}^{(1)} + \cdots + c_n\vec{x}^{(n)} = \Psi(t)\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = \Psi(t)\vec{c}, \vec{c} \in \mathbb{R}^n \text{ is arbitrary.}$$

2). $\det \Psi(t) = W\left(\vec{x}^{(1)}, \cdots, \vec{x}^{(n)}\right)(t) \neq 0, \quad t \in I \Rightarrow \Psi(t)$ *is invertible.*

3). Fundamental matrix is not unique because fundamental set of solution is not unique.

Theorem

(i). Consider $\vec{x}' = P(t)\vec{x}$. *Let* $\Psi(t)$ *be any fundamental matrix and* A *be any nonsingular constant matrix, then* $\Psi(t)A$ *is also a fundamental matrix.*

(ii). Consider the IVP: $\begin{cases} \vec{x}' = P(t)\vec{x}, \\ \vec{x}(t_0) = \vec{x}^0 \end{cases}$ *.Let* $\Psi(t)$ *be any fundamental matrix,then* $\Phi(t) = \Psi(t)\Psi^{-1}(t_0)$ *is also a fundamental matrix,and the unique solution to the IVP is* $\vec{x} = \Phi(t_0)\vec{x}^0 = \Psi(t)\Psi^{-1}(t_0)\vec{x}^0$.

Remark

For (i) : in general, $A\Psi(t)$ *is not fundamental matrix.*
For (ii): There are infinitely many fundamental matrix $\Psi(t)$*, however,* $\Phi(t) = \Psi(t)\Psi^{-1}(t_0)$ *is unique for fixed* t_0*, because* $\Phi(t)$ *is the unique solution to the IVP*

$$\begin{cases} \frac{d\Phi}{dt} = P(t)\Phi, \\ \Phi(t_0) = I_n \end{cases}$$

Matrix exponential

Key idea for this section:
For single equation

$$\begin{cases} x' = ax, & a \in \mathbb{R}. \\ x(0) = x^0, \end{cases}$$

Solution is $x(t) = e^{at}x^0$.
Expect for system

$$\begin{cases} \vec{x}' = A\vec{x}, & A = n \times n \text{ constant matrix}, \\ \vec{x}(0) = \vec{x}_0, \end{cases} \qquad \text{solution is } \vec{x}(t) = e^{At}\vec{x}_0$$

Definition

We define matrix exponential using Taylor series:
$e^A = \sum_{m=0}^{+\infty} \frac{1}{m!}A^m = I + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \cdots$ is a $n \times n$ constant matrix.

$$e^{At} = \sum_{m=0}^{+\infty} \frac{1}{m!}A^m t^m \text{ is a } n \times n \text{ matrix function of } t.$$

Properties of matrix exponential

Theorem

Let A, B be $n \times n$ constant matrices. Then

i). $e^0 = I_n$, *where* $0 = \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ldots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$

ii). If $AB = BA$, *then* $e^{(A+B)} = e^A e^B$

iii). For any matrix A (may be invertible or not), e^A is always invertible, and $\left(e^A\right)^{-1} = e^{-A}$.

iv). For any matrix A and any invertible matrix P,

$$e^{PAP^{-1}} = Pe^A P^{-1}.$$

Diagonalizable matrix

Consider $\vec{x}' = A\vec{x}$. From the theory of linear algebra, A is diagonalizable $\Leftrightarrow$ there exist n linearly independent eigenvectors $\vec{\xi}_1, \cdots, \vec{\xi}_n$ corresponding to eigenvalues $\lambda_1, \cdots, \lambda_n$ (may be equal).
In this case, set $P = \left(\vec{\xi}_1, \cdots, \vec{\xi}_n\right)$,

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

the fundamental matrix $\Phi(t)$ is

$$\Phi(t) = e^{tA} = P \begin{pmatrix} e^{\lambda_1 t}t & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{pmatrix} P^{-1}.$$

And we can choose another fundamental matrix $\Psi(t)$:

$$\Psi(t) = \Phi(t)P = P \begin{pmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{pmatrix} = \left(e^{\lambda_1 t}\vec{\xi_1}, \cdots, e^{\lambda_n t}\vec{\xi_n} \right),$$

Example

Let $A = \begin{pmatrix} 5 & 4 \\ -8 & -7 \end{pmatrix}$. Show that A is diagonalizable and compute e^A.

Sol: $\det(A - \lambda I) = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = -3$.

For $\lambda_1 = 1$, $\vec{\xi_1} = \binom{-1}{1}$, for $\lambda_2 = -3$, $\vec{\xi_2} = \binom{-1}{2}$. Let

$P = \left(\vec{\xi_1}, \vec{\xi_2}\right) = \begin{pmatrix} -1 & -1 \\ 1 & 2 \end{pmatrix}$, then $P^{-1}AP = \begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix}$.

$$e^A = P \begin{pmatrix} e & \\ & e^{-3} \end{pmatrix} P^{-1} = \begin{pmatrix} 2e - e^{-3} & e - e^{-3} \\ -2e + 2e^{-3} & -e + 2e^{-3} \end{pmatrix}.$$

Jordan decomposition

How to compute e^{At} if A is not diagonalizable?

Theorem (A important theory from linear algebra)

If λ is an eigenvalue of A with algebraic multiplicity m, then $(A-\lambda I)^m v = 0$ has m linearly independent solutions: $\{\vec{v}_1, \vec{v}_2, \cdots, \vec{v}_m\}$ They are called generalized eigenvector.
Let $P = (v_1|v_2|\ldots|v_n)$ consists of all generalized eigenvectors, we can write $P^{-1}AP = J$, where P is made up of the eigenvectors and generalized eigenvectors,

$$J=\begin{pmatrix} J_1 & 0 & \ldots & 0 \\ 0 & J_2 & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & J_n \end{pmatrix} \quad ,J_i = \begin{pmatrix} \lambda_i & 1 & 0 & \ldots & 0 \\ 0 & \lambda_i & 1 & \ldots & 0 \\ 0 & 0 & \lambda_i & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ 0 & 0 & \ldots & \lambda_i & 1 \\ 0 & 0 & \ldots & 0 & \lambda_i \end{pmatrix}$$

Jordan decomposition

For the exponential, this becomes

$$e^J = \begin{pmatrix} \exp(\lambda) & \exp(\lambda) & \cdots & \frac{\exp(\lambda)}{(k-1)!} \\ & \exp(\lambda) & \ddots & \vdots \\ & & \ddots & \exp(\lambda) \\ & & & \exp(\lambda) \end{pmatrix}$$

Hence we have:

$$e^{Jt} = \begin{pmatrix} \exp(\lambda)t & \exp(\lambda)t^2 & \cdots & \frac{\exp(\lambda)}{(k-1)!}t^k \\ & \exp(\lambda)t & \ddots & \vdots \\ & & \ddots & \exp(\lambda)t^2 \\ & & & \exp(\lambda)t \end{pmatrix}$$

You will learn more about this in MAT3040.

Exercise

Find the foundamental matrix for:

(i)

$$x' = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} x$$

(ii)

$$x' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 8 & -5 & 3 \end{pmatrix} x$$

Exercise

Find the foundamental matrix for:

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(ii)

$$x' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -8 & -5 & -3 \end{pmatrix} x$$

Exercise

(i)

$$x' = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} x$$

Sol: We have $\begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} = PDP^{-1}$ where $P = \begin{pmatrix} -1 & 1 \\ 4 & 1 \end{pmatrix}, D = \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix}$

Hence $\Psi(t) = \begin{pmatrix} -e^{-3t} & e^{2t} \\ 4e^{-3t} & e^{2t} \end{pmatrix}$

(ii) Similarly we can solve $\Psi(t) = \begin{pmatrix} -4e^{-2t} & 0 & -3e^{-t} \\ 5e^{-2t} & e^{2t} & 4e^{-t} \\ 7e^{-2t} & -e^{2t} & 2e^{-t} \end{pmatrix}$

Nonhomogeneous Linear System

Consider

$$\vec{x}' = P(t)\vec{x} + \vec{g}(t)$$

General solution is

$$\vec{x} = c_1\vec{x}^{(1)} + \cdots c_n\vec{x}^{(n)} + \vec{v}(t)$$

where $\vec{x}^{(1)}, \cdots, \vec{x}^{(n)}$ are fundamental set of solutions to $\vec{x}' = P(t)\vec{x}$, $\vec{v}(t)$ is a particular solution to $\vec{x}' = P(t)\vec{x} + \vec{g}(t)$.

Question: How to find $\vec{v}(t)$?
Method of undetermined coefficients
Variation of parameter

Method of undetermined coefficients

Works for: $\vec{x}' = A\vec{x} + \vec{g}_{(t)}$.

$$\begin{cases} A \text{ is a constant matrix,} \\ \vec{g}_{(t)} \text{ is sum or product of } \vec{p}_n(t), e^{at}, \cos\beta t, \sin\beta t. \end{cases}$$

$\vec{P}_n(t)$, vector valued polynomial of degree n : $\vec{a}_n t^n + \vec{a}_{n-1} t^{n-1} + \cdots + \vec{a}_1 t + \vec{a}_0$.

$\vec{g}(t)$	$\vec{v}(t)$	s
$\vec{P}_m(t)$	$\vec{Q}_{m+s}(t)$	multiplicity of 0
$\vec{P}_m(t)e^{\alpha t}$	$\vec{Q}_{m+s}(t)e^{\alpha t}$	multiplicity of α
$\vec{P}_m(t)e^{\alpha t}\cos\beta t$ or $\vec{P}_m(t)e^{\alpha t}\sin\beta t$	$(\vec{Q}_{m+s}(t)\cos\beta t + \vec{R}_{m+s}(t)\sin\beta t)e^{\alpha t}$	multiplicity of $\alpha + i\beta$

Remark: Compare

$$\begin{cases} n\text{th order equation: the particular solution has the form } t^s Q_m(t), \\ \text{system: the solution is expected to be } Q_{m+s}(t). \end{cases}$$

Example

Find a particular solution to $\vec{x}' = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}\vec{x} + \binom{4e^{-t}}{18t}$.

Sol: Let $\vec{g}(t) = e^{-t}\binom{4}{0} + t\binom{0}{18} = \vec{g}_1(t) + \vec{g}_2(t)$.

$\det(A - \lambda I) = (\lambda + 3)(\lambda + 1) = 0, \lambda_1 = -3, \lambda_2 = -1$.

For $g_1(t) = e^{-t}\binom{4}{0}, -1$ is a root with multiplicity $s = 1$.

$\Rightarrow V_1(t) = \vec{Q}_1(t)e^{-t} = (\vec{a}t + \vec{b})e^{-t}$.

For $g_2(t) = t\binom{0}{18}, 0$ is not a root, $s = 0$. $\Rightarrow V_2(t) = \vec{R}_1(t) = \vec{c}t + \vec{d}$.

Set $v(t) = (\vec{a}t + \vec{b})e^{-t} + \vec{c}t + \vec{d}$.Substituted to the system,

$\Rightarrow -\vec{a}te^{-t} + (\vec{a} - \vec{b})e^{-t} + \vec{c} = A\vec{a}te^{-t} + A\vec{b}e^{-t} + A\vec{c}t + A\vec{d} + \vec{g}$

Compare the coefficients for te^{-t}, e^{-t}, t, t^0, $\Rightarrow \begin{cases} (A + I)\vec{a} = 0. \\ (A + I)\vec{b} = \vec{a} - \binom{4}{0}, \\ A\vec{c} = -\binom{0}{18}. \\ A\vec{d} = \vec{c}. \end{cases}$

We can find one solution (Not unique):

$\vec{a} = \binom{2}{2}, \vec{b} = \binom{0}{-2}, \quad \vec{c} = \binom{6}{12}, \vec{d} = \binom{-8}{-10}$. so $\vec{v}(t) = \binom{2te^{-t}+6t-8}{2te^{-t}-2e^{-t}+12t-10}$.

Exercise

Write the form for the particular solution of the following:

$$\vec{x}' = \begin{pmatrix} 1 & 5 \\ -1 & -1 \end{pmatrix} \vec{x} + \begin{pmatrix} e^{2t} \\ \sin 2t \end{pmatrix}.$$

$$\vec{x}' = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \vec{x} + \begin{pmatrix} e^{3t} \\ 6e^{t} \end{pmatrix}$$

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Sol:

$$\vec{v}(t) = e^{2t}\vec{a} + \cos 2t(t\vec{b} + \vec{c}) + \sin 2t(t\vec{d} + \vec{e})$$

$$\vec{v} = e^{3t}\vec{a} + t^2 e^{t}\vec{b} + te^{t}\vec{c} + e^{t}\vec{d}$$

Variation of parameter

Works for general system $\vec{x}' = P(t)\vec{x} + \vec{g}_{(t)}$.
If $\Psi(t)$ is a fundamental matrix for the homogeneous system, then assume the particular solution to the nonhomogeneous system is

$$\begin{aligned}
&\vec{x}(t) = \Psi(t)\vec{c}(t).\\
\Rightarrow \quad &\Psi'(t)\vec{c}(t) + \Psi(t)\vec{c}'(t) = P(t)\Psi(t)\vec{c}(t) + \vec{g}(t)\\
\Rightarrow \quad &\Psi(t)\vec{c}'(t) = \vec{g}(t)\\
&\vec{c}(t) = \int \Psi^{-1}(t)g(t)dt
\end{aligned}$$

Particular solution is $\Psi(t) \cdot \int \Psi^{-1}(t)g(t)dt$
General solution is $\Psi(t)\vec{c} + \Psi(t) \cdot \int \Psi^{-1}(t)g(t)dt, \vec{c} \in \mathbb{R}^n$.

Example

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Sol: $\lambda_1 = -3, \quad A + 3I = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \vec{\xi}_1 = \binom{1}{-1}, \quad \vec{x}^{(1)} = e^{-3t}\binom{1}{-1}$.

$\lambda_2 = -1, \quad A + I = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \vec{\xi}_2 = \binom{1}{1}, \vec{x}^{(2)} = e^{-t}\binom{1}{1}$.

so $\Psi(t) = \begin{pmatrix} e^{-3t} & e^{-t} \\ -e^{-3t} & e^{-t} \end{pmatrix}, \quad \Psi^{-1}(t) = \frac{1}{2}\begin{pmatrix} e^{3t} & -e^{3t} \\ e^{t} & e^{t} \end{pmatrix}$

$\int \Psi^{-1}(t)g(t)dt = \int \binom{2e^{2t}-9te^{3t}}{2+9te^{t}} dt = \binom{e^{2t}-3te^{3t}+e^{3t}}{2t+9te^{t}-9e^{t}}$.

So a particular solution is

$$\vec{V}(t) = \Psi(t) \int \Psi^{-1}(t)g(t)dt$$

$$= \begin{pmatrix} e^{-3t} & e^{-t} \\ -e^{-3t} & e^{-t} \end{pmatrix} \begin{pmatrix} e^{2t} - 3te^{3t} + e^{3t} \\ 2t + 9te^{t} - 9e^{t} \end{pmatrix}.$$

$$= \begin{pmatrix} 2te^{-t} + e^{-t} + 6t - 8 \\ 26e^{-t} - e^{-t} + 12t - 10 \end{pmatrix}.$$

Exercise

$$\vec{x}' = \begin{pmatrix} 4 & -2 \\ 8 & -4 \end{pmatrix} \vec{x} + \begin{pmatrix} t^{-3} \\ -t^{-2} \end{pmatrix}$$

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Sol:

$$\vec{x} = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} t \\ 2t - \frac{1}{2} \end{pmatrix} - \frac{1}{t^2} \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} + \frac{1}{t} \begin{pmatrix} 2 \\ 5 \end{pmatrix} - 2 \ln t \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

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Sol:

$$\vec{v}(t) = e^{2t}\vec{a} + \cos 2t(t\vec{b} + \vec{c}) + \sin 2t(t\vec{d} + \vec{e})$$

$$\vec{v} = e^{3t}\vec{a} + t^2 e^{t}\vec{b} + te^{t}\vec{c} + e^{t}\vec{d}$$

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Particular solution is $\Psi(t) \cdot \int \Psi^{-1}(t)g(t)dt$
General solution is $\Psi(t)\vec{c} + \Psi(t) \cdot \int \Psi^{-1}(t)g(t)dt, \vec{c} \in \mathbb{R}^n$.

Example

$\vec{x}' = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \vec{x} + \binom{4e^{-t}}{18t}$.

Sol: $\lambda_1 = -3, \quad A + 3I = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \vec{\xi}_1 = \binom{1}{-1}, \quad \vec{x}^{(1)} = e^{-3t}\binom{1}{-1}$.

$\lambda_2 = -1, \quad A + I = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \vec{\xi}_2 = \binom{1}{1}, \vec{x}^{(2)} = e^{-t}\binom{1}{1}$.

so $\Psi(t) = \begin{pmatrix} e^{-3t} & e^{-t} \\ -e^{-3t} & e^{-t} \end{pmatrix}, \quad \Psi^{-1}(t) = \frac{1}{2}\begin{pmatrix} e^{3t} & -e^{3t} \\ e^{t} & e^{t} \end{pmatrix}$

$\int \Psi^{-1}(t)g(t)dt = \int \binom{2e^{2t}-9te^{3t}}{2+9te^{t}} dt = \binom{e^{2t}-3te^{3t}+e^{3t}}{2t+9te^{t}-9e^{t}}$.

So a particular solution is

$$\vec{V}(t) = \Psi(t) \int \Psi^{-1}(t)g(t)dt$$

$$= \begin{pmatrix} e^{-3t} & e^{-t} \\ -e^{-3t} & e^{-t} \end{pmatrix} \begin{pmatrix} e^{2t} - 3te^{3t} + e^{3t} \\ 2t + 9te^{t} - 9e^{t} \end{pmatrix}.$$

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Exercise

$$\vec{x}' = \begin{pmatrix} 4 & -2 \\ 8 & -4 \end{pmatrix} \vec{x} + \begin{pmatrix} t^{-3} \\ -t^{-2} \end{pmatrix}$$

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$$\vec{x} = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} t \\ 2t - \frac{1}{2} \end{pmatrix} - \frac{1}{t^2} \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} + \frac{1}{t} \begin{pmatrix} 2 \\ 5 \end{pmatrix} - 2 \ln t \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$